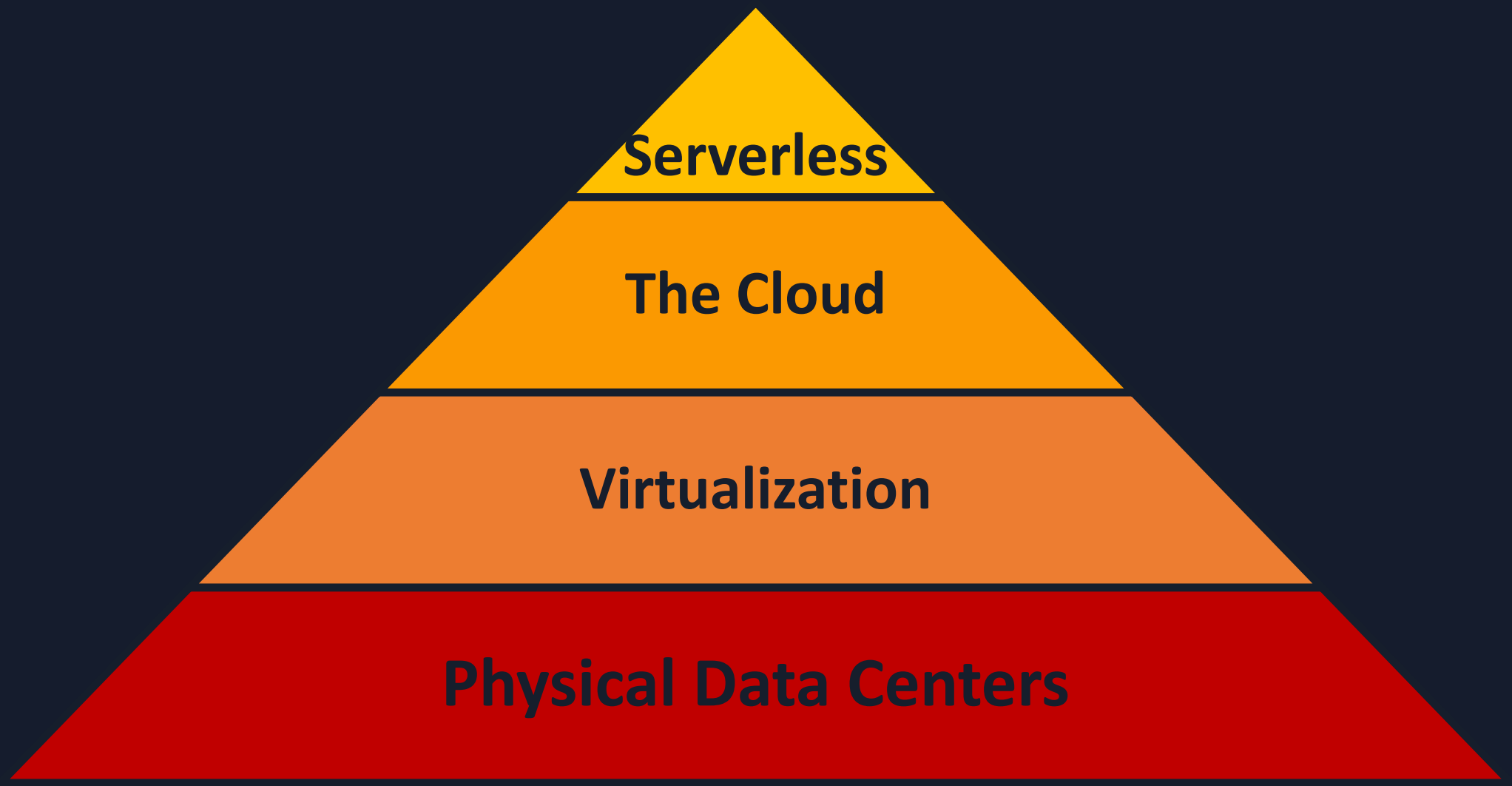
Amazon Web Services: Serverless

Topics

- Serverless Overview
- Computing with Lambda
- Containers
- Running Containers in Amazon ECS or EKS
- Removing Servers with Fargate

Serverless Overview

History of Computing



Serverless Benefits

Ease of Use

AWS handles almost everything for us. We just bring our code

Event Based

Serverless compute resources can be brought online in response to an event happening

Billing Model

“Pay as you go” in its purest form. You only pay for the provisioned resources and the length of runtime

What’s Coming Up

AWS Lambda

AWS Fargate

Computing with Lambda

What is AWS Lambda?

- Serverless compute service that lets you run code without provisioning or managing the underlying servers
- No more overhead; focus on code
- Allows you to run you code on-demand with the added benefit of automated scaling
- Host and run code in Lambda functions

Lambda Features

Pricing

- Free tier of 1,000,000 requests and 400,000 GBs of compute per month; after that, pay per request

Integration

- Integrates with numerous AWS services including S3, DynamoDB, EventBridge, SQS/SNS, and Kinesis

Built-In Monitoring

- Logging and monitoring can be easily accomplished with CloudWatch

Easy Configuration

- Easily set memory requirements needed. Currently able to use up to 10,240 MB. CPU scales with memory

Execution Length

- Used for short-term executions. Time limit of 900 seconds (15 minutes). For anything longer, use ECS, Batch, or EC2

Runtime Support

- Leverage industry-dominant languages including Python, Golang, Java, Node.js, and others

Building a Lambda Function

Configuration

Runtime: Pick from an available runtime or bring your own. This is the environment code will run in

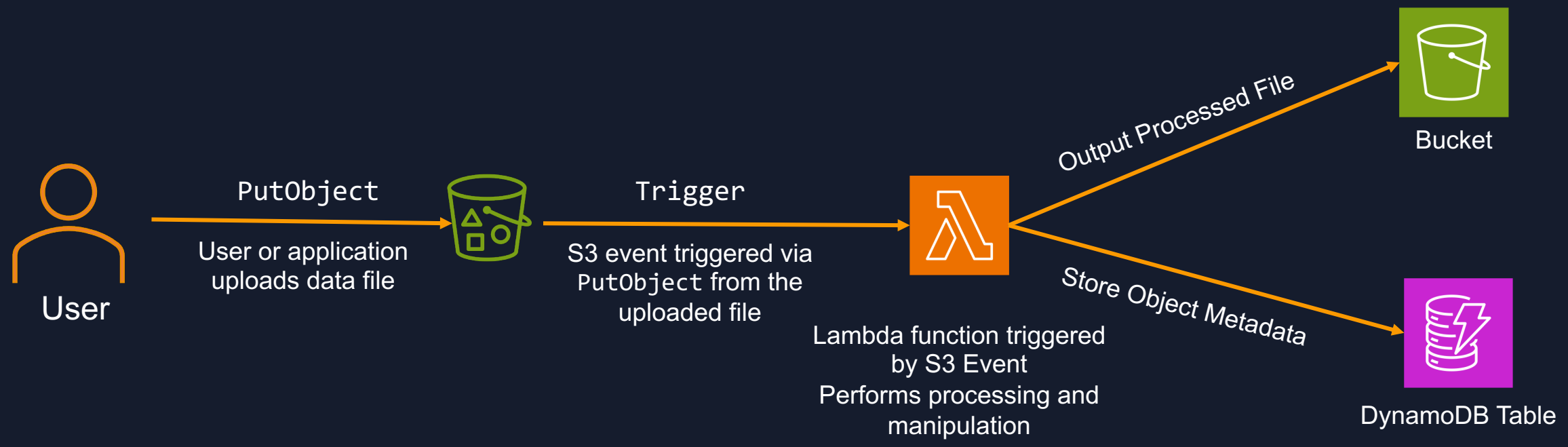
Permissions: If your Lambda function needs to make an AWS API call, you'll need to attach a role

Networking: Optionally define the VPC, subnet and security groups your functions are a part of

Resources: Define the amount of available memory allocated to your function

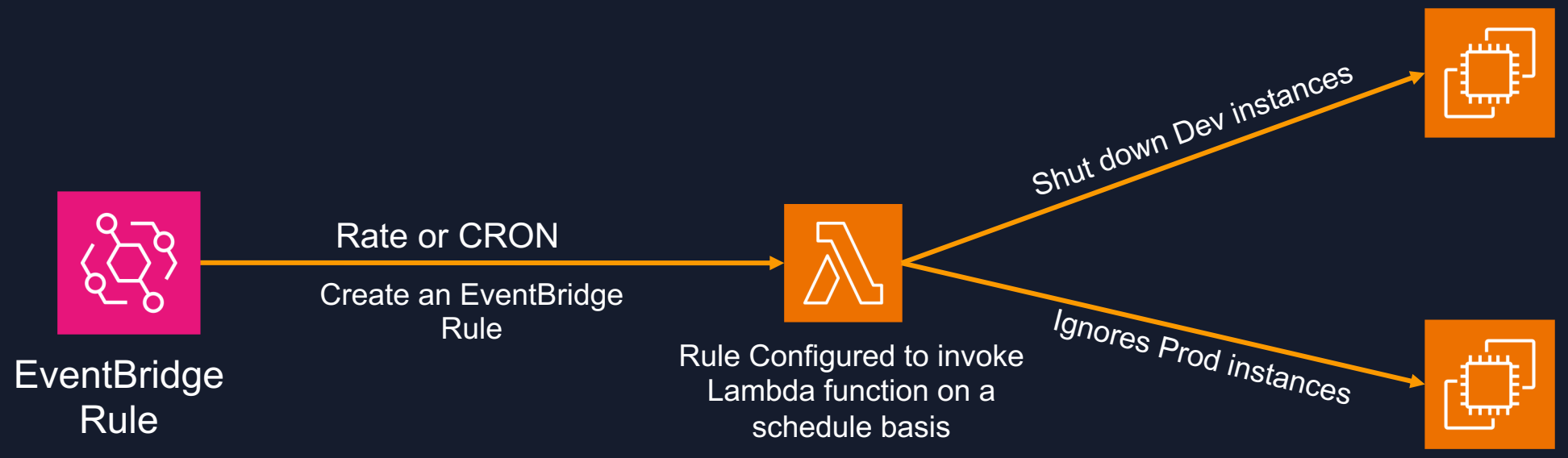
Triggers: What's going to alert your Lambda function to start? Defining a trigger will kick Lambda off if that event occurs

Popular Architecture Examples



Lambda can integrate with other AWS services via IAM Role

Popular Architecture Examples



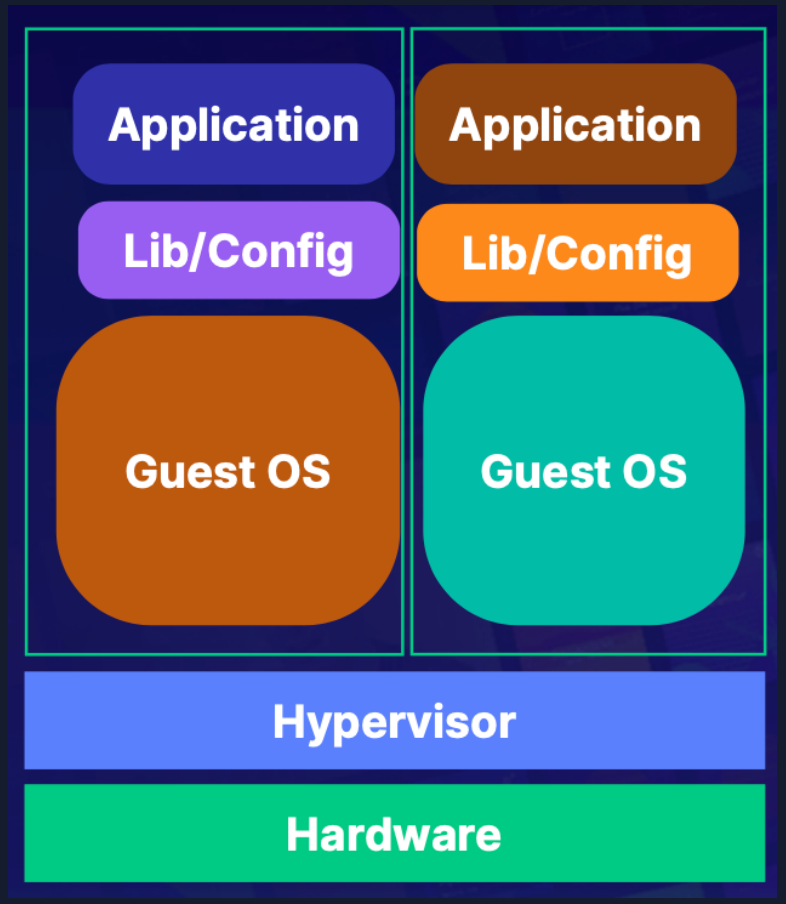
Lambda Demo

Containers

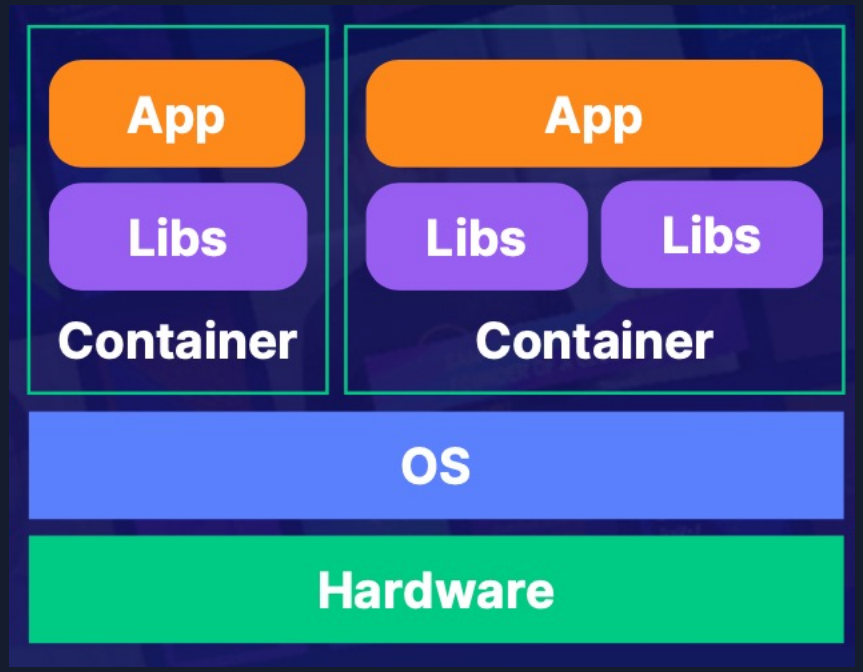
What is a Container?

- A container is a standard unit of software that packages up code and all its dependencies, so the application runs quickly and reliably from one computing environment to another

Virtual Machine vs. Container



Virtual Machine



Container

Container Terminology

Dockerfile

Text document that contains all the commands or instructions that will be used to build an image

Image

Immutable file that contains the code, libraries, dependencies, and configuration files needed to run an application

Registry

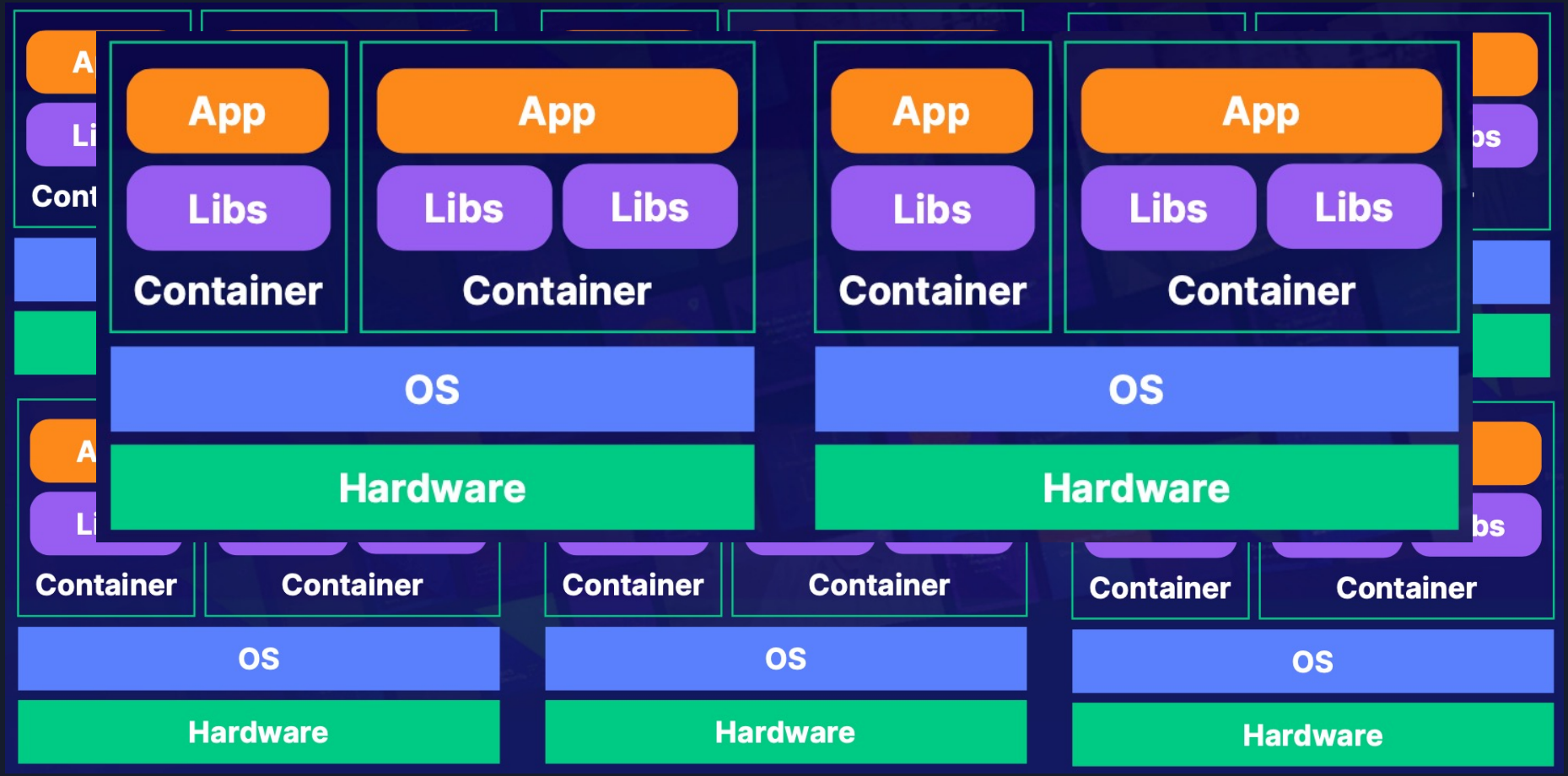
Stores Docker images for distribution. They can be both private and public

Container

A running copy of the image that has been created

Running Containers in Amazon ECS or EKS

Problems with Containers



Amazon ECS

- ECS can manage from one to thousands of containers
- ECS will appropriately place the containers and keep them online
- Containers are appropriately registered with the load balancers as they come online and deregistered as they go offline
- Containers can have individual roles attached to them, making security a breeze
- Extremely easy to set up and scale to handle any workload

Amazon Elastic Container Service (Amazon ECS)

Other Container Orchestration

- Kubernetes
  - Open-source
  - AWS managed version EKS

Amazon Elastic Kubernetes Service (Amazon EKS)

ECS vs. EKS

ECS

- Proprietary AWS container management solution
- Best used when you're all in on AWS
- Looking for something simple

VS

EKS

- AWS-managed version of open-source Kubernetes container management solution
- Best used when you're not all in on AWS
- More work to configure and integrate with AWS

Removing Servers with Fargate

What is Fargate?

- AWS Fargate is a serverless compute engine for containers
- AWS owns and manages the underlying infrastructure
- Requires use of ECS or EKS
- Can use both Linux and Windows containers

ECS Launch Types: EC2 vs. Fargate

EC2

- You are responsible for underlying operating system
- EC2 pricing model vs
- Long-running containers
- Multiple containers share the same host
- Capable of mounting EFS file systems for persistent, shared storage

VS

Fargate

- No operating system access
- Pay based on resources allocated and time ran
- Short-running tasks
- Isolated environments
- Capable of mounting EFS file systems for persistent, shared storage

What about Lambda?

Fargate

- Select Fargate when you have more consistent workloads
- Allows Docker use across the organization and a greater level of control by developers

Lambda

- Great for unpredictable or inconsistent workloads
- Perfect for applications that can be expressed as a single function

Demo: Spin up a new task in ECS using Fargate Launch Type

Checkpoint

What is the maximum length of time Lambda can run?

1. 30 minutes
2. 15 minutes
3. 1 hour
4. 1 minute

What is one thing EC2 instances allow you to configure but a serverless application doesn't?

1. Operating System
2. The ability to pay for the service.
3. VPC placement
4. The ability to configure the service.

What type of file is used to build a container image?

1. Image file
2. Dockerfile
3. Container definition
4. build_info.txt

What feature of ECS and EKS allows you to run containers without having to manage the underlying hosts?

1. Fargate
2. Lambda
3. EC2
4. S3

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